NHS Primary Care Dementia Analysis

Abstract

This report analyses dementia data in the UK, especially for people aged 65 and over. Using NHS-provided datasets, the report presents two visualisations created with Tableau Public. The first visualisation explores gender-based dementia cases, showing that women are more affected than men, particularly in the 75–79 age group. The second visualisation focuses on estimated diagnosis rates and dementia types across Integrated Care Boards (ICBs). The most common dementia type is Alzheimer's disease, followed by other types and vascular dementia. The analysis also highlights regional differences in diagnosis performance.

Data preprocessing was a key part of the project, as some records were unstructured or incomplete. Tools like Excel Power Pivot cleaned and prepared the data before joining datasets for analysis. The findings help identify gaps in diagnosis and offer valuable insights for improving dementia care. This report will inform NHS Directors and support better planning, resource allocation, and early intervention strategies. It shows how data visualisation can help make healthcare decisions more effective and targeted.

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Introduction

Dementia is a significant global health concern, affecting 47.5 million people globally, including 850,000 in the United Kingdom, and is anticipated to approach 2 million by 2050. Public Health England (PHE) wants to lower the prevalence of dementia by encouraging people to live healthily in middle age. Dementia, which is commonly associated with brain illnesses, causes memory loss as well as issues with thinking and communicating. It is the most dreaded health issue among those over 65 (NHS, 2023), surpassing cancer and heart disease. Women are disproportionately impacted, both as healthcare workers and patients, and they frequently experience loneliness and sadness. Ethnic minorities, notably Black and South Asian groups, are at a higher risk because of underlying health issues. Without significant public health action, the number of dementia cases will continue to climb dramatically (Public Health England, 2021). The number of people with dementia is rising as life expectancy increases. By 2030, it is expected that over 1 million people in the UK will be living with dementia (NHS, 2023).

According to Gokalp (2025), this report used links in the assignment remit to find relevant Primary Care Dementia Data sets from the NHS for additional research (NHS England, 2024). The NHS (National Health Service) is the UK's publicly funded healthcare system that provides medical services to all UK citizens without requiring payment at the time of care. These services include doctor and nurse visits, hospital treatments, maternity care, and emergency assistance, which are all paid for by taxes (Full Fact, 2017). This report includes two research studies: one is aimed at the general public and is designed for a news audience, while the other is intended for NHS Directors to support internal planning, improve services, and highlight key healthcare challenges using data-driven insights (Gokalp, 2025).

Early detection is an effective technique for avoiding and controlling serious illnesses, including dementia. Many diseases, particularly in their early stages, produce no symptoms and need regular health tests. Detecting diseases early improves survival rates, enables less intrusive treatments, improves quality of life, and lowers healthcare costs. For example, early-stage cancer treatment is significantly more successful and less demanding than late-stage treatment. The NHS provides various screening programs, including cardiovascular health, diabetes, and dementia evaluations, essential in detecting disorders before they worsen. Encouraging early intervention and raising public awareness are crucial steps toward decreasing the long-term burden of illnesses like dementia (Echelon Health, 2025).

These data visualisations effectively convey critical results to various target groups. The first visualisation focuses on gender-based trends in dementia data, whereas the second visualisation focuses on persons aged 65 and older who have been diagnosed with dementia. Tableau Public was used to showcase these findings in an understandable and interactive format. The links to both visualisations have been provided in their respective report sections (Gokalp, 2025).

Dementia by Gender Overview

For the Public News Audience visualisation, this research uses the Primary Care Dementia Data, September 2024: Cognitive Impairment dataset provided by the NHS Organisation (NHS England, 2024). This study aims to investigate how dementia affects different genders throughout age ranges, especially those aged 40 to 90 or beyond. This visualisation assists in determining if men or women are more typically impacted at specific periods of life. The purpose of studying age and gender patterns is to increase public awareness and enhance early detection initiatives. The visualisation is intended for a public news audience and seeks to convey the findings in an easily understood and accessible manner using Tableau Public.

Tableau:
https://public.tableau.com/views/1main/Public?:language=en-US&publish=yes&:sid=&:redirect=auth&:display_count=n&:origin=viz_share_link

Datasets:
https://drive.google.com/drive/folders/1eoLmYZOsC6gvCEcHmh0XOGhnhaoNaK-s?usp=sharing

The dataset was initially unclean, with the "measure" column containing unclear values like *MCI_GENDER_AGED_X_Y*. Using Power Pivot and a delimiter, this column was split into two separate fields: Gender and Age Range (X_Y). This cleaning step made it easier to perform further analysis based on gender and age groups. A bar chart was created using Tableau Public to display the number of dementia patients for each month between June and September 2024. The bar chart shows that dementia cases grew monthly throughout this time frame. In each month, there were more female dementia cases than male cases. Most of these incidents were recorded in regions covered by the Sub-Integrated Care Board (Sub ICB) (NHS England, 2023). All age groups saw increased cases, although those aged 75 to 79 had the most patients each month. A tree map was also used to depict various sorts of healthcare organisations, such as Sub ICBs, Integrated Care Boards (ICBs) (NHS England, 2023), and Commissioning Regions (NHS England, 2023). The tree map showed that Sub ICB locations had the most cases in all months and age groups. Females accounted for 54% of all reported instances at each organisation. A tooltip feature was added to display the top 10 care boards with the highest number of dementia patients, assisting in determining which locations were most affected.

According to the pie chart, males account for 46% of dementia cases and females account for 54%. Both genders saw identical monthly case patterns, which the same organisations reported. The majority of instances for both males and females were between the ages of 75 and 79, with the fewest in the 40-44 range. The bubble graphic shows that females had greater case counts across most age groups, particularly at 90+, accounting for 65%. However, men had significantly greater numbers in several 70-79 age groups.

In our analysis, it is clear that females have a higher chance of being affected by dementia than males. This finding aligns with other research, showing that two out of every three people with dementia are female, and twice as many women die from it, likely due to their longer life expectancy (Dementia UK, 2025). Research shows that about 4 in 10 dementia cases could be

prevented by managing changeable risk factors. Experts say that what benefits the heart also helps the brain. Key steps include eating a balanced diet, regular exercise, maintaining a healthy weight, limiting alcohol, quitting smoking, and controlling blood pressure (NHS, 2023).

Estimated Dementia Rates in Older Adults (65+)

Dementia is a syndrome that involves a decline in brain function, affecting memory, thinking, language, emotion, and conduct. It is not a typical aspect of ageing, although the risk grows with age, particularly for individuals over 65. Memory loss, slow thinking, poor judgment, confusion, difficulty with daily duties, and personality changes are some of the symptoms. People may have trouble communicating and socialising, as well as experiencing hallucinations. The most frequent types of dementia are Alzheimer's and vascular dementia. As dementia progresses, individuals frequently require assistance with daily activities and decision-making. In the United Kingdom, more than 944,000 people suffer from dementia, with one in every 11 persons aged 65 or older affected (NHS, 2023).

Since dementia is not a standard component of ageing, and the risk increases dramatically with age, particularly for people over 65, a relevant dataset titled “Primary Care Dementia Data, September 2024: Summary” (NHS England, 2024) this was used for this analysis. Worksheet 4 of this dataset contains the 65+ estimated dementia diagnosis rate, which explains how widely dementia is diagnosed among older adults in various regions. The data was paired with additional data on “Primary Care Dementia Data, September 2024: Dementia type by Sub ICB Location” (NHS England, 2024), allowing researchers to determine which forms are most typically registered across various care boards. This combined research provides a more complete picture of the prevalence and type of dementia in the 65+ population across healthcare organisations.

Before joining the datasets, data preprocessing was a crucial step in this analysis to ensure consistency, accuracy, and usability. One of the most challenging tasks was changing the Types of Dementia dataset, which included several types of dementia such as Alzheimer's disease, frontotemporal dementia, inconclusive dementia type, Lewy body dementia, mixed dementia types, other dementia types, and vascular dementia. Using Python, these dementia types were initially reported as values in a single column, which made direct analysis difficult. The dataset was flipped to prepare the data for meaningful visualisation so that each dementia kind became its column, with the appropriate values assigned. This change enabled a more accurate comparison and analysis of how dementia kinds are distributed across various care boards.

The primary dataset titled "Summary" contained estimated dementia diagnosis rates, but only for September 2024, and some records had missing ODS codes. Any unmatched or old records were eliminated to ensure data consistency, including entries from earlier months in the dementia types dataset. The cleaned summary dataset contained crucial fields including ODS Code, ONS Code, Name, Recorded, Estimated, Diagnosis Rate (%), Lower, and Upper. I did an inner join between the two datasets, utilising the ONS Code from the summary table and the ICB ONS Code from the dementia types dataset. The final combined data is uniquely tailored to the ICB (Integrated Care Board) organisation type. After this join, the resulting

dataset contained **42 rows and 15 columns**, representing the matched and usable records for analysis and visualisation in the second part of the project.

Tableau:
https://public.tableau.com/views/2Main/NHS?:language=en-US&publish=yes&:sid=&:redirect=auth&:display_count=n&:origin=viz_share_link

Data sets:
https://drive.google.com/drive/folders/1EPZszsIbNDKgPrUpppIeOPmiaXGXN3HA?usp=sharing

The histogram depicts how many care boards fall into various dementia diagnosis rate ranges, focusing on patients aged 65 and older. Most care boards (11 in total) have diagnostic rates ranging from 60% to 62%. This implies that numerous care boards diagnose dementia at the same rate, indicating a consistent level of identification and recording techniques across multiple regions. In contrast, fewer care boards fall into ranges above 76% or below 53%, indicating that very high or very low diagnostic rates are uncommon. To make the chart more informative, a tooltip feature was added. This enables users to rapidly display the top three Organisation Data Service (ODS) codes with the most estimated dementia patients within the specified diagnosis range. In this example, the top three ODS codes are QRL (17,084 patients), QKS (15,535 patients), and QU9 (14,626 patients). This feature provides easy reference points and helps determine which care boards contribute the most to that range, making the visualisation more interactive and helpful for further analysis.

The bullet graph shows the top 10 care boards with the highest estimated dementia cases among those 65 and over. Unlike a regular bar chart, the bullet graph includes reference lines to compare estimated and recorded dementia figures. NHS North East and North Cumbria Integrated Care Board has the highest estimate with 41,154 cases, followed by NHS Cheshire and Merseyside (34,733) and NHS Greater Manchester (30,489). Other care boards, such as NHS Sussex, West Yorkshire, and Hampshire, also report high cases ranging from approximately 23,000 to 28,000. This visual helps identify the regions most affected by dementia and highlights where additional support or targeted intervention may be needed. The reference lines in the graph also make it easy to see the gap between estimated and recorded cases, which can guide future planning and improve diagnosis efforts.

The scatter plot illustrates the association between the projected number of dementia patients aged 65 and up and the diagnosis rate across various Integrated Care Boards (ICBs). Each bubble represents an ICB, with the colour corresponding to the ICB name and the bubble size indicating the higher diagnostic rate. The x-axis depicts the estimated number of patients, while the y-axis represents the related diagnosis rate. A trend line has been added to indicate the overall direction of the data, revealing a little upward slope, implying a weak positive link between the number of patients and the diagnostic rate.

The R-squared value is 0.0226, which means that only about 2.26% of the variation in diagnosis rate can be explained by the number of estimated patients. The p-value is 0.342, indicating that the relationship is not statistically significant. This suggests that the number of estimated

patients does not strongly influence how well dementia is diagnosed across care boards. The chart helps identify which areas need improved screening or awareness and gives a clear view of regional performance differences using size and colour distinctions.

The bar chart displays the estimated number of cases for different types of dementia recorded across various care boards. Alzheimer’s disease is the most common, with 115,465 estimated instances, followed by Other Dementia Types with 86,620 cases, and Vascular Dementia with 42,990 cases. Mixed Dementia Types also show a notable count of 18,550, while Lewy Body dementia stands at 8,810 cases. Less common types include Inconclusive Dementia Type with 4,515 cases and Frontotemporal dementia with just 1,945 cases—the lowest among all categories. This visualisation helps identify which types of dementia are most prevalent and may require more focused medical attention, research, or resources within the healthcare system. It also supports a better understanding of how dementia is classified and tracked across regions.

This report provides a detailed picture of dementia diagnostic trends and distribution among care boards, focusing on people aged 65 and over. The findings show significant heterogeneity in diagnosis rates, with some care boards outperforming others. Alzheimer's disease is still the most common kind, followed by other and vascular dementias, emphasising critical areas for specialised care and budget allocation. Although there is an upward trend between estimated cases and diagnosis rates, the association is weak, implying that factors other than patient volume influence diagnosis accuracy. Understanding these patterns allows NHS Directors to identify gaps, prioritise high-need regions, and strengthen early detection and intervention strategies, ultimately improving patient outcomes and care efficiency.

Key Findings and Discussion

The dataset provides a comprehensive perspective of dementia cases across many NHS Integrated Care Boards for adults aged 65 and up. It featured essential columns like recorded and estimated cases, diagnosis rates, and various kinds of dementia. The statistics revealed that some locations have high diagnosis rates, while others remain below the national norm. This implies that there may be variations in screening techniques, public awareness, or access to diagnostic services. The bullet graph highlighted which care boards had the most estimated patients, which is essential for more equitable resource planning and distribution.

In addition to the overall figures, a breakdown of dementia types provides extra context for the analysis. Alzheimer's disease remains the most prevalent, although a considerable proportion of patients are classified as "Other Dementia Types" or "Vascular Dementia." The range of categories indicates that care providers must be adaptable and informed about various symptoms and demands. Some dementia types, such as Lewy Body and Frontotemporal, are less prevalent but present in the dataset and should not be ignored. Overall, these data assist NHS Directors in determining where services are operating effectively and where additional support or improvements are required to guarantee equitable and timely treatment for all seniors.

Conclusion

This research demonstrates the importance of using actual data to understand better and enhance dementia treatment within the NHS. Using visual tools, we identified critical regions with high diagnostic rates and others with opportunities for improvement. However, the data is more than simply statistics; it represents the requirements of older individuals living with dementia and the systems that support them. As the UK's older population grows, so will the demand for early diagnosis, adequate treatment, and long-term care. NHS decision-makers must take these developments seriously and prepare to deliver improved patient care and carer support.

One of the issues encountered while working with the dataset was the lack of well-structured data. Some entries lacked critical data, such as entire ODS codes or dementia kinds, and not all organisations provided values for every field. This made it challenging to examine the data appropriately initially. Cleaning and preparing the data was a time-consuming procedure to obtain trustworthy results. If the NHS or data providers can enhance data collection and sharing, analysis will become more easier and more accurate. This would also allow care boards to take quicker action when needed.

To enhance dementia services, more effort should be placed on staff training, better use of technology, and raising awareness. AI-based screening, digital health data, and automated alarms can help clinicians detect dementia earlier and more accurately. Public education efforts should also inform people, particularly those in their midlife, how to reduce risk by adopting healthier living habits. For example, encouraging more nutritious diets, regular exercise, and blood pressure management can all help prevent dementia cases in the long run.

Finally, fixing these issues will need collaboration. Local care boards, national NHS leaders, and other health groups must collaborate. Sharing knowledge and working towards common goals can help guarantee that all dementia patients, regardless of where they live, receive equitable and high-quality treatment. This research demonstrates that with the appropriate data and tools, we can make better decisions to help persons living with dementia and develop a stronger healthcare system for the future.

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